



GE Medical Systems

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Technical Publication

Direction 2373637-100

Revision 01

GE Medical Systems

Octane1 1.0GB Memory Upgrade

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
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- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

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14 July 1993

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All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

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Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

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End of Section

Revision History

Revision	Date	Reason for change
0	03/24/03	Initial Release
1	04/22/03	Updated "System Requirement" section.

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Octane1 1.0GB Memory Upgrade

Section 1.0 System Requirement

Octane1 single processors that need to load (or have recently loaded) FMI 25314.

Section 2.0 CPU Memory Upgrade Installation Preparation

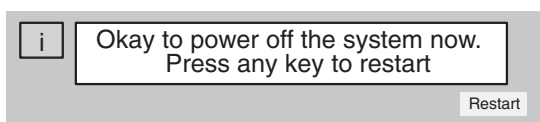
The CPU Memory Upgrade adds an additional 1.0 GB of CPU memory to the Silicon Graphics Computer. This additional memory allows more efficient execution of system software by reducing the amount of disk swap activity and allowing larger amounts of data to be available in memory at any given moment.

2.1 Shutdown the System

- 1.) Select the (square red) **SHUTDOWN** button (shown below) on the Image Display Monitor. Select **OK** to shutdown Applications. This will stop both Applications and UNIX software and bring you to the boot prompt.



- 2.) After the software shuts down, the following message will appear.



- 3.) Turn off the power switch on the console.

2.2 Preparing the Workstation to Install the Optional Memory



NOTICE



THE MEMORY CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.

Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.

To prepare the workstation for installation of optional memory, follow these steps:

- 1.) Remove the front cover of the Console.
- 2.) Unplug the power cord from the electrical outlet and from the OCTANE workstation.

 **CAUTION** The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.

- 3.) Wait 5 minutes before removing the system module.
- 4.) If applicable, remove the lockbar that locks the system module and bezel to the chassis.
 - Unlock the lock at the back of the OCTANE workstation.
 - Pull the lockbar out of the OCTANE workstation.

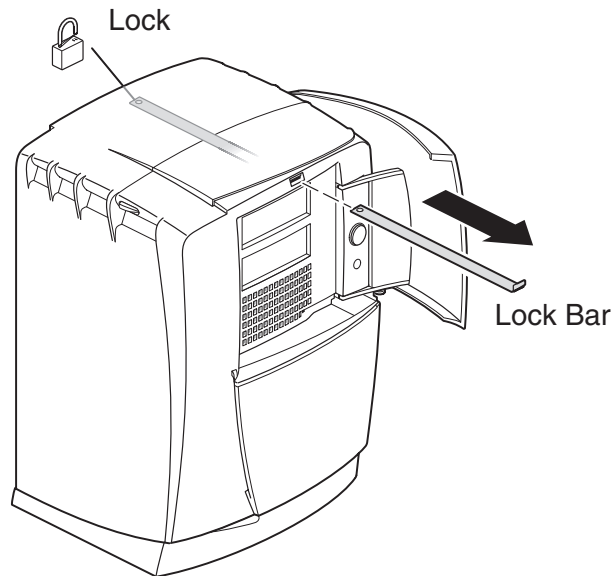


Figure 1 Lock Bar Removal (If Applicable)

- 5.) Face the rear of the workstation.
- 6.) Remove the cables attached to the system module.

2.3 Removing the System Module



NOTICE



THE CIRCUIT BOARDS ARE VERY STATIC-SENSITIVE.

Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards.



CAUTION

The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.

- 1.) To remove the system module, you need to power off the system, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, and attach the wrist strap.
- 2.) Loosen the captive screws that hold the sliding handles to the OCTANE workstation.

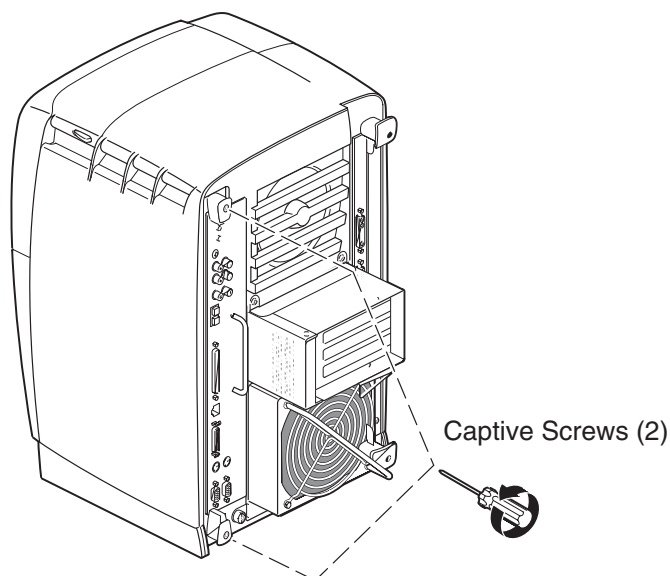


Figure 2 Removing the System Module Screws

Look for the two compression connectors when you take the system module out of the chassis, and do not touch them. They are on the back of the system module, and connect to the chassis when the system module is installed in the workstation.



NOTICE

The gold bristled pad on the compression connectors on the back of the system module is a very delicate and easily damaged bristled pad.

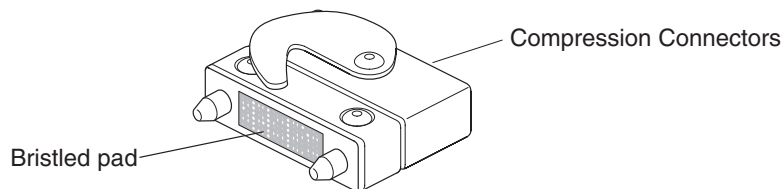


Figure 3 Locating the Compression Connectors

- 3.) Pull both sliding handles simultaneously until they are completely extended. This action releases the system module from the workstation.

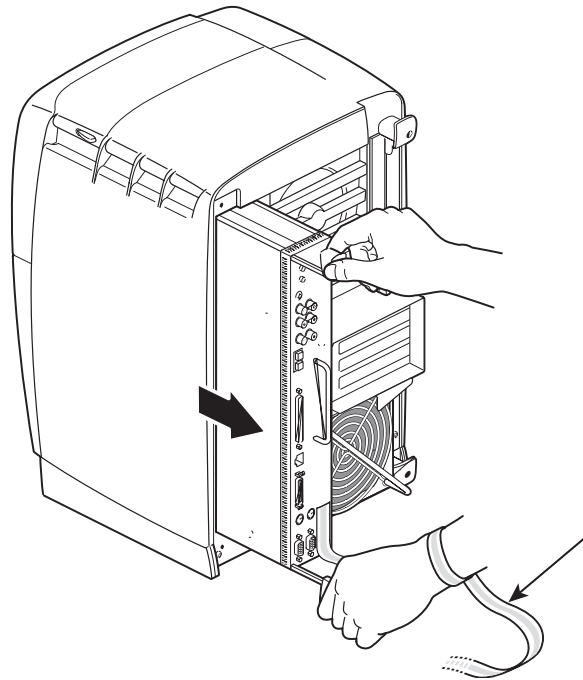


Figure 4 Releasing the System Module from the Front Plane

- 4.) Pull the system module from the chassis by grasping the module handle with your left hand and bracing your right hand against the top of the workstation. Support the bottom of the module as you pull it out.
- 5.) Place it on a dry, antistatic surface with the CPU and DIMMs facing up. A desktop works well.



NOTICE

Do not touch the compression connectors as you remove the system module. The side of the compression connector that connects to the workstation is extremely delicate.

- 6.) Place a cap on each compression connector on the back of the system module. (Compression connector caps come with the system.)



CAUTION

Warning: The heat sinks on the CPU and system module become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the system module. Test before touching the CPU or heat sinks.

Section 3.0

Installing and Removing Memory

To install or remove memory, you must power off the workstation, wait 5 minutes after powering off the workstation to allow the heat sinks to cool, attach the wrist strap, and remove the system module.



NOTICE



DIMMS ARE VERY STATIC-SENSITIVE.

Static-sensitive components may be damaged if not handled in a static-free environment. Take appropriate precautions (e.g., wear properly grounded wrist strap) when handling all boards. Dual In-line Memory Modules (DIMMs) are extremely sensitive to static electricity. Handle the DIMMs carefully, and wear the wrist strap to avoid the effects of static electricity.

3.1 About Memory

Before you install DIMMs in your OCTANE workstation, read the following information.

- Bank 1 (sockets 1 and 2) must always be filled.
 - Minimum memory configuration is 64 MB in a bank. (2 x 32 MB DIMMs)
 - Maximum configuration is 512 MB in a bank. (2 x 256 MB DIMMs)
- Banks are filled sequentially; when Bank 1 is full, fill Bank 2. Do not skip banks.
- By common practice, if all of the memory installed is not of the same capacity, it is always a good idea to populate the DIMM Banks closest to the processor with the highest available capacity DIMMs. (i.e. Bank 1 contains the highest capacity DIMMs, bank 2 contains the same capacity DIMMs or the next lower capacity, and so on for banks 3 & 4.
- Each bank is either empty, or contains two DIMMs, one in each of the two sockets.
- Each bank of DIMMS must contain two DIMMs of the same capacity or density and of the same type.
 - Capacity refers to the number of megabytes in a DIMM: 32, 64, 128, 256, and so on.
 - Type refers to the construction of the DIMM.

Because DIMM construction cannot be determined visually, it is important to know that two DIMMs are the same type before you begin installation. If you insert two DIMMs of different types, the system will not work. Banked DIMMs and stacked DIMMs are two examples of many different types of memory.
 - It is always safest to ensure that both DIMMs in a bank are from the same manufacturer.
- Minimum memory capacity in the OCTANE workstation is 64 MB. (2 x 32 MB DIMMs)
- Maximum memory capacity in the OCTANE workstation is 2 GB. (8 x 256 MB DIMMs)

3.2 Installing Memory

Note: This option kit consists of 4 x 256 MB DIMMs (1GB), which will be installed in the lowest numbered Memory Banks (Banks 1 & 2). Your system will already have at least 512 MB of memory installed. This may consist of either 2 x 256 MB DIMMs, or 4 x 128 MB DIMMs. If your system has additional memory (beyond 512 MB) installed (such as the standard 3-D Option memory), you may have to remove and discard it to install the 4 x 256 MB DIMMs included in this kit.

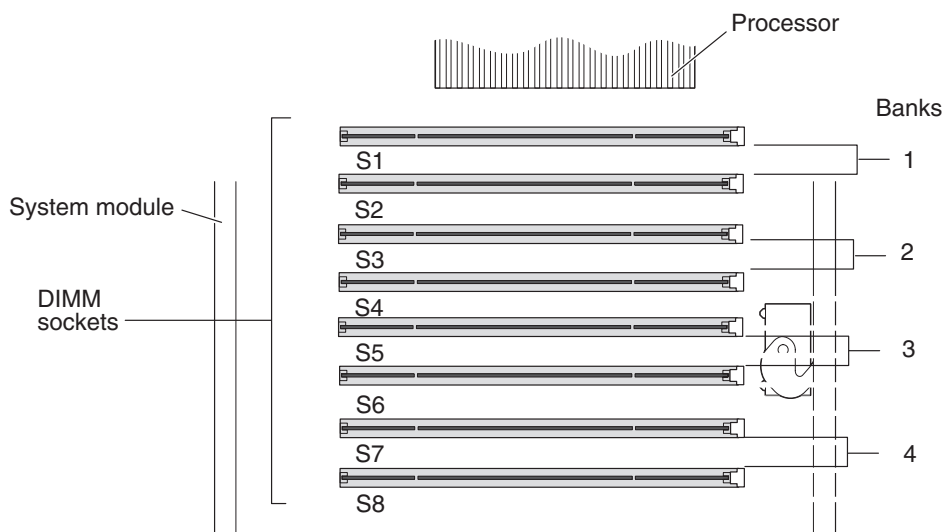


Figure 5 Identifying DIMM Sockets and DIMM Banks

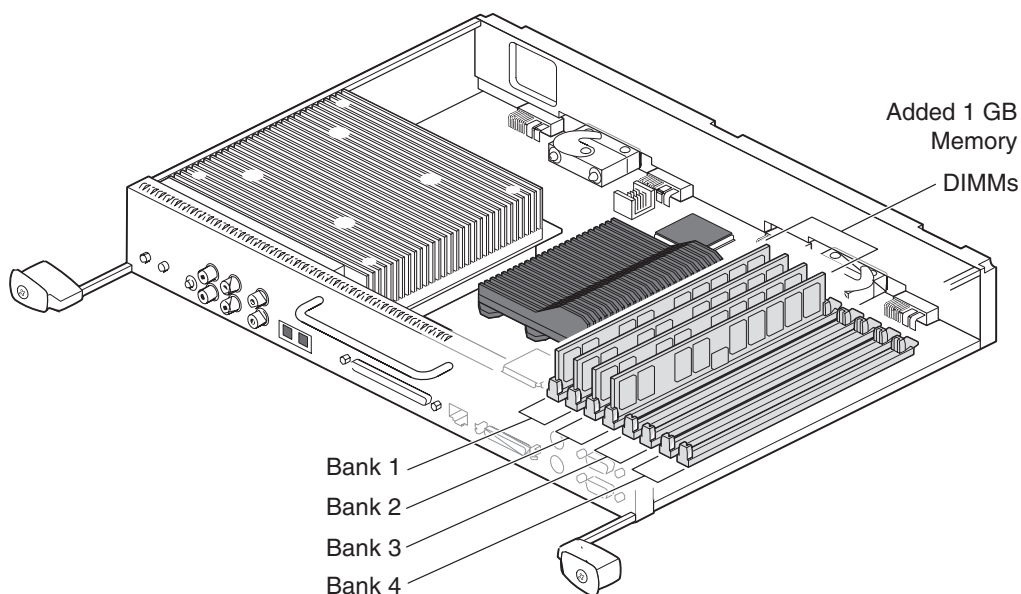


Figure 6 Locating DIMMs on the System Module

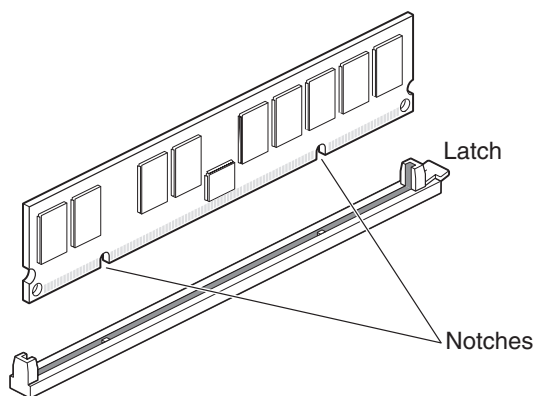


Figure 7 Locating the Notches on a DIMM

DIMMs are notched on the bottom so that they cannot be inserted incorrectly (see [Figure 7](#)).

- 1.) Remove all existing DIMMs from all sockets, and set them aside.
- 2.) Using the DIMMs provided in this kit, insert each DIMM into the socket gently but firmly, starting with slot location S1, and continuing through slot S4 (Banks 1 & 2). You hear a click as the DIMM is seated, and the latch on the end of the socket moves up.
- 3.) Check to be sure both sockets in each bank are full. DIMMs must be installed in pairs.
- 4.) Install all 4 x 256 MB DIMMs provided in this kit into Banks 1 & 2. Refer to section 3.1 on page 17.
- 5.) Re-install the removed DIMMs (maximum of 4) into the remaining memory sockets (S5 through S8) of Banks 3 & 4. If these DIMMs are of different capacities (or if you aren't sure), see [Table 1, "DIMM Identification," on page 19](#) for information about DIMM identification, and review the information about socket location vs. DIMM capacity in section 3.1 on page 17.
- 6.) Once you have finished installing the memory, you are ready to replace the system module.

DIMM Manufacturer / Memory Capacity	Manufacturer Part Number	Other Coding (i.e. label color)	Comments
SGI - 32MB	9940069	Yellow	Never used on QX/i
Dataram - 32MB	60056	n/a	Never used on QX/i
SGI - 64MB	9940084 9470178	Blue Green	Used on standard 3-D Option only
Dataram - 64MB	62614	n/a	Used on standard 3-D Option only
SGI - 128MB	9470168	Brown*	Standard Memory
SGI - 128MB	9010020	Brown**	Standard Memory
Dataram - 128MB	62615	n/a	Standard Memory
SGI - 256MB	9010021	White***	Standard/Direct-3D
Dataram - 256MB	62621	High Profile	Direct-3D Option
Dataram - 256MB	62649	Low Profile	Direct-3D Option

Notes:

* 2K Refresh DIMM's, must be used as a matched pair in a bank, typically used in the original IP-30 System Module (SGI P/N 030-0887-003).

** 4K Refresh DIMMs, must be used as a matched pair in a bank, and will only function in an enhanced IP-30 System Module (SGI P/N 030-1467-001).

*** 4K Refresh DIMMs, must be used as a matched pair in a bank, and will only function in an enhanced IP-30 System Module (SGI P/N 030-1467-001).

Table 1 DIMM Identification

3.3 Replacing the System Module

- 1.) Remove the caps on the compression connectors on the back of the system module. Do not touch the system module compression connectors as you lift the module into the workstation.

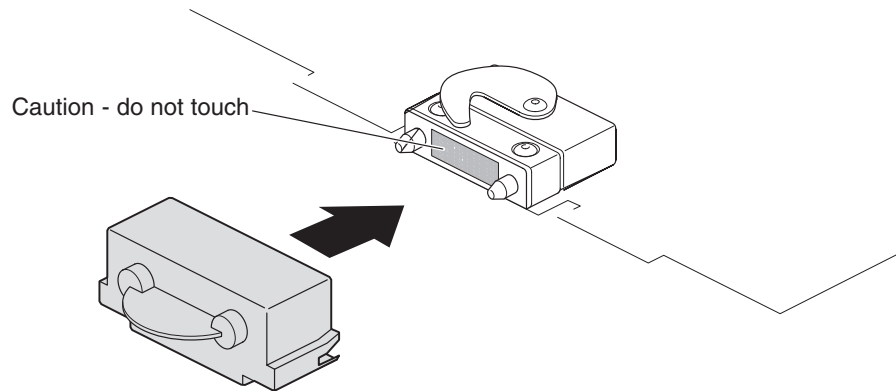


Figure 8 Removing the Compression Connector Cap



NOTICE

Do not insert the system module into the workstation unless the power cord is unplugged from the electrical socket. Inserting the system module into an OCTANE workstation with the power cord connected to an electrical socket could cause damage to the OCTANE workstation.

- 2.) Grasp the system module by its immovable handle and support it with one hand as you slide it into the chassis.

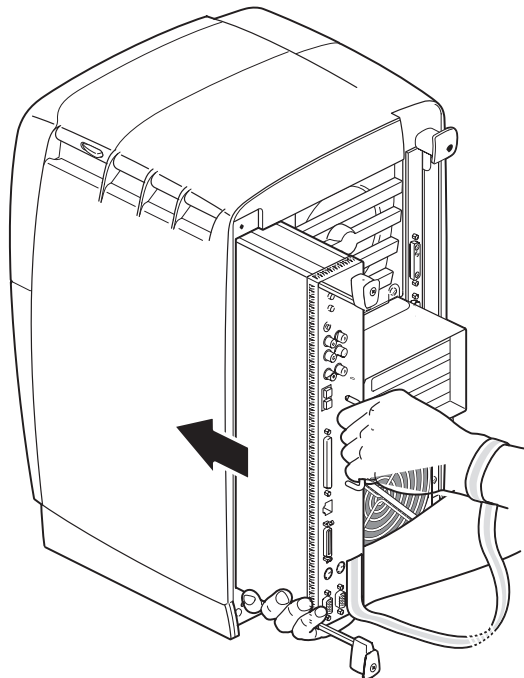


Figure 9 Supporting the System Module

- 3.) Push the module completely into the chassis using the immovable handle. (Both sliding handles protrude.) The system module stops when it is slightly out of the chassis.

- 4.) Push the sliding handles simultaneously to connect the system module to the workstation.

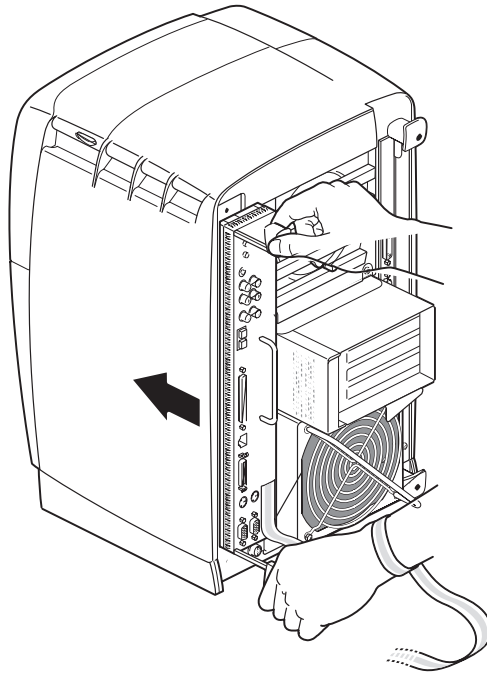


Figure 10 Replacing the System Module in the OCTANE Workstation

- 5.) Push firmly on the handles to completely lock the system module to the workstation.
The handles are completely recessed and the system module is flush with the chassis when it is fully seated and locked to the workstation.
- 6.) Tighten the captive screws in the sliding handles until the system module is attached to the chassis.

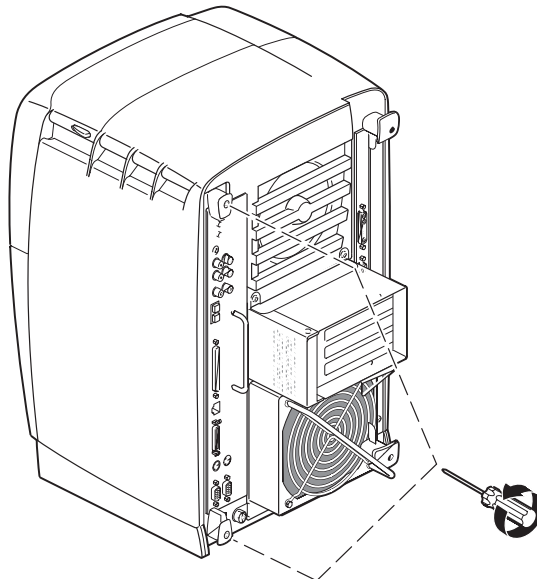


Figure 11 Replacing the System Module Screws

- 7.) Remove the wrist strap.

3.4 Powering On the OCTANE Workstation

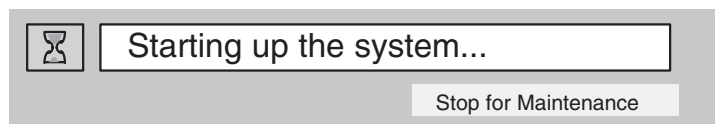
- 1.) Plug in the power cord.
- 2.) Replace the lockbar (if applicable).
 - Open the cover on the front of the OCTANE workstation.
 - Slide the lockbar into the lockbar slot.
 - Insert the lock through the lockbar at the back of the workstation.

Section 4.0

Start Up System and Verify Installed Memory

4.1 Start System

- 1.) Turn on the console power switch.
- 2.) The following message will appear. Select the STOP FOR MAINTENANCE button.



4.2 Verify Installed Memory

The system Maintenance menu appears.

- 1.) Select ENTER COMMAND MONITOR.
- 2.) At the shell prompt type: `hinv`
- 3.) Verify that the Main memory size is at least 1.5 GB (1512 MB).

Note: If the amount of memory is incorrect, follow the directions at the beginning of this chapter to power off your system and check the installation procedure.

- Check the angle of the DIMMs. They should be upright and completely seated.
- Check that each bank is populated with two DIMMs, and that they are of the same type. You must have an even number of DIMMs installed.

- 4.) Exit the shell by typing: `exit`
- 5.) Select START SYSTEM

The system software will start normally.

- 6.) After the system is started, open a shell tool and re-check the hardware environment:

```
{ctuser@somesystem} [1] hinv
1 XXX MHZ IP30 Processor
CPU: MIPS RXXXXX Processor Chip Revision: 2.7
FPU: MIPS RXXXXX Floating Point Chip Revision: 0.0
Main Memory size: 1512 Mbytes    <- This is the minimum memory you should see.
Instruction cache size: 32 Kbytes
Data cache size: 32 Kbytes
Secondary unified instruction/data cache size: 1 Mbytes
Integral SCSI controller 0: Version QL1040B (rev. 2)
  Disk drive: unit 1 on SCSI controller 0
Integral SCSI controller 1: Version QL1040B (rev. 2)
  Optical disk: unit 4 on SCSI controller 1
  Comm device: unit 4 on SCSI controller 1
  Comm device: unit 4, lin 1 on SCSI controller 1
  CDROM: unit 6 on SCSI controller 1
IOC3 serial port: tty1
IOC3 serial port: tty2
IOC3 parallel port: plp1
Graphics board: ESI with texture option
Graphics board: ESI
Integral Fast Ethernet: ef0, version 1, pci 1
Fast Ethernet: ef1, version 1, pci 3
Iris Audio Processor: version RAD revision 12.0, number 1
```




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